Eugene Vuong

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San Jose, CA

Research Interests

Ph.D. student in the Systems and Networking Lab (SNL) at UC Santa Barbara, advised by Prof. Arpit Gupta. My research focuses on machine learning for networking and systems, AI agents for automating network application workflows, realistic network traffic dataset generation, and large-scale Internet and broadband measurement to inform policy.

Experience

- **Lawrence Berkeley National Laboratory** *Oct. 2026 - Present*
  Graduate Student Research Assistant (UC GSRA) Berkeley, CA, USA
  - Will conduct research advised by Prof. Arpit Gupta and the ESnet group at Lawrence Berkeley National Laboratory

- **University of California, Santa Barbara** *Feb. 2026 - Aug. 2026*
  Staff Research Associate, Systems and Networking Lab Santa Barbara, CA, USA
  - Scaled the NetGent AI-agent framework to autonomously navigate to pricing pages across 100+ ISP domains using natural-language instructions, supporting the BQT+ broadband measurement framework [S.2]
  - Engineered a web scraping pipeline using RabbitMQ and Docker to orchestrate task queuing, managing 100% of user-submitted jobs with high-availability parallel processing
  - Led the design and launch of a full-stack Next.js SaaS application for robust and extensible measurement of 1,000+ broadband plans, managing four undergraduates and coordinating with another PhD student

- **University of California, Santa Barbara** *Jun. 2025 - Aug. 2025*
  Research Intern, Systems and Networking Lab Santa Barbara, CA, USA
  - Developed NetGent, an AI-agent framework that automates complex web workflows to generate realistic network traffic datasets for advancing machine learning in networking [W.1]
  - Validated the framework across 50+ workflows, cutting the projected LLM cost of a million-run data collection from $98,000 to under $10 in annual maintenance
  - Optimized the data collection pipeline with a Huey task queue orchestrating 50 concurrent Docker containers, reducing average workflow time by over 57%

- **Lawrence Berkeley National Laboratory** *Jun. 2023 - Aug. 2023*
  Research Intern Berkeley, CA, USA
  - Enhanced a PyTorch U-Net architecture for elemental analysis with residual connections, decreasing Root Mean Square Error (RMSE) by 25% relative to the original U-Net
  - Accelerated emission spectrum simulation by 96% using Numba
  - Improved data integrity in a 118,000-record dataset using Pandas, refining approximately 59.5% of entries

- **California State University, East Bay** *Sep. 2022 - Aug. 2024*
  Data Analyst Intern Hayward, CA, USA
  - Used Python to identify and correct user-ID errors in the Blackbaud Raiser's Edge CRM, improving data accuracy by 11.18% across 91,000+ records
  - Implemented a PyAutoGUI automation script to update 7,400+ entries, reducing update time by 30% compared to manual methods
  - Reduced the magazine mailing list by 38.04% across 50,000 records by standardizing addresses, de-duplicating at person/household levels, and suppressing invalid/opt-out contacts with Pandas, cutting print and postage spend

Education

- **University of California, Santa Barbara** *Sep. 2026 - Present (Expected Jul. 2031)*
  Ph.D. in Computer Science Santa Barbara, CA, USA
  - Advisor: Prof. Arpit Gupta, Systems and Networking Lab (SNL)

- **California State University, East Bay** *Dec. 2025*
  B.S. in Computer Science Hayward, CA, USA
  - GPA: 4.00/4.00
  - Coursework: Computer Networks, Operating Systems, Computer Architecture, Data Structures and Algorithms, Programming Languages, Natural Language Processing, Software Engineering, Web Development, Discrete Structures

PROJECTS

- **EasyAds: Autonomous AI Agents for Video Ad Generation** *Self-Evolving Agents Hack 2025*
  Tools: Flask, Next.js, Linkup, Freepik, Lovable [🌐]
  - Won Best Use of Freepik's API at the Self-Evolving Agents Hack with a multi-agent platform that finds trending products and generates video ads for them
  - Orchestrated Linkup for real-time product trend discovery and Freepik for visual asset sourcing within a multi-agent pipeline
  - Generated video ads in under 5 minutes using caching and parallel agent processing
- **Mentora: AI Agent that Turns Any Prompt into a Learning Roadmap** *Cal Hacks 2025*
  Tools: Letta, Flask, Firebase, Next.js, OpenAI [🌐]
  - Won Honorable Mention (Letta Track) among 50+ teams at Cal Hacks (2,000+ participants)
  - Built an asynchronous Flask API with background job processing and polling endpoints, increasing throughput under load by offloading long-running tasks
  - Designed a multi-agent recursive tree-expansion system in which a RootAgent seeds high-level topics and search-enabled NodeAgents iteratively expand and enrich each branch
- **Second Sight: AI Security System with Semantic Video Search** *Multimodal Hackathon 2025*
  Tools: Activeloop Deep Lake, FastAPI, Next.js, Together AI, Google Gemini, Firebase [GitHub]
  - Won 1st place (Activeloop Sponsor Prize Track) among 150+ teams with a system that repurposes legacy devices into smart cameras with semantic video search
  - Integrated Google Gemini and Together AI into a motion detection pipeline to generate captions and vector embeddings for natural-language footage retrieval
  - Stored captions and embeddings in Activeloop Deep Lake for RAG, enabling sub-second natural-language search over 10+ hours of footage
- **CallFlow: AI Customer Service Assistant** *AI Tinkerers Hackathon 2024*
  Tools: Next.js, TailwindCSS, Google Gemini, Flask [🌐]
  - Programmed an AI customer service assistant to automate routine tasks and escalate complex actions for human approval, winning 1st place (Grand Prize) at the AI Tinkerers Humans-in-the-Loop Agents Hackathon
  - Utilized Google Gemini and Vapi to build a voice AI agent with fraud detection and human escalation, ensuring trustworthy AI-customer interactions
  - Designed an intuitive UI with Next.js and TailwindCSS for real-time monitoring, enabling oversight of AI decisions

PUBLICATIONS

C=CONFERENCE, W=WORKSHOP, S=IN SUBMISSION / PREPRINT

[C.1] **A Study of Large Language Model Performance for Phishing URL Detection and Explainability for User-Centric Security**. Jaishnoor Kaur, **Eugene Vuong**, Hongmin Li. In *2026 Silicon Valley Cybersecurity Conference (SVCC)*, pp. 1-8. IEEE. San Jose, CA, USA, 2026. DOI: 10.1109/SVCC69905.2026.11642313.

[W.2] **Pramana: A Composable, Domain-Specific Backend for Empirical Networking Research**. Jaber Daneshamooz, **Eugene Vuong**, Alagappan Ramanathan, Manni Moghimi, Haarika Manda, Satyam Kumar, Snithik Thode, Satyandra Guthula, Sylee Beltiukov, Dongsu Han, Tarun Mangla, Sangeetha Abdu Jyothi, Walter Willinger, Arpit Gupta. In *25th ACM Workshop on Hot Topics in Networks (HotNets 2026)*, Salt Lake City, UT, USA, 2026. arXiv:2607.26352.

[W.1] **NetGent: Agent-Based Automation of Network Application Workflows**. Jaber Daneshamooz, **Eugene Vuong**, Laasya Koduru, Sanjay Chandrasekaran, Arpit Gupta. In *NeurIPS 2025 Workshop on Machine Learning for Systems (ML4Sys)*, 2025. arXiv:2509.00625.

[S.2] **Robust and Extensible Measurement of Broadband Plans with BQT+**. Laasya Koduru, Sylee Beltiukov, Alexander Nguyen, **Eugene Vuong**, Jaber Daneshamooz, Tejas Narechania, Elizabeth Belding, Arpit Gupta. *arXiv preprint* arXiv:2602.16969, 2026.

[S.1] **Enabling Data-Driven Policymaking Using Broadband-Plan Querying Tool (BQT+)**. Laasya Koduru, Sylee Beltiukov, Jaber Daneshamooz, **Eugene Vuong**, Arpit Gupta, Elizabeth Belding, Tejas N. Narechania. *arXiv preprint* arXiv:2511.05838, 2025.

SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, Java, C++, HTML, CSS
- **Web Technologies:** React.js, Next.js, Express.js, Flask, FastAPI
- **Database Systems:** PostgreSQL, MongoDB, Firebase
- **AI, Data Science & Machine Learning:** PyTorch, LangChain, LangGraph, Letta, Pandas, NumPy, Matplotlib, BeautifulSoup4
- **Cloud Technologies:** AWS, GCP
- **DevOps & Infrastructure:** Docker, Kubernetes, Git, RabbitMQ, Apache Airflow
- **Research Skills:** AI agents for web automation, network traffic dataset generation, broadband and Internet measurement, large-scale data collection pipelines

Honors and Awards

- **CRA Outstanding Undergraduate Researcher Award, Honorable Mention** *Dec. 2025*
  Computing Research Association (CRA) [🌐]
  - Recognized as an Honorable Mention in the 2025–26 CRA Outstanding Undergraduate Researcher Award program
  - The award recognizes undergraduates at North American colleges and universities who show outstanding potential in computing research, based on faculty nominations
- **Cal-Bridge Fellowship (Cohort 11)** *Jun. 2024*
  Cal-Bridge Program [🌐]
  - Selected for a CSU–UC pathway program that prepares CSU students in computer science, physics/astronomy, and mathematics to pursue PhDs, with over 350 scholars across its 11 cohorts
  - Received financial support, joint mentoring from CSU and UC faculty, professional development workshops, and research opportunities at UC campuses
- **Ytha and Joanne Yu Scholarship** *2024*
  California State University, East Bay [🌐]
  - Endowed Computer Science scholarship awarded on the basis of past academic performance and future academic promise
  - Recognized at the CSU East Bay Mathematics and Computer Science 2023–2024 awards presentation

Leadership Experience

- **Lead** *Aug. 2024 - May 2025*
  Google Developer Group, California State University, East Bay Hayward, CA, USA
  - Led the CSU East Bay chapter of Google Developer Group, growing the student developer community to 400+ members
  - Organized regular meetups and workshops to give students hands-on, industry-relevant experience alongside a theory-heavy CS curriculum, open to all experience and skill levels
  - Invited an AI Software Engineer from GripTape AI to introduce their LLM framework to members
  - Directed a core team, delegating venue logistics to the Operations Lead, visual materials to the Design Lead, and a physical and digital marketing campaign to the Vice President